

STRATA: A Stratified Process-Mining Framework for HVAC Fault Detection and Diagnosis

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Abstract

Faults in building HVAC equipment persist because the tools meant to catch them either raise alarms that operators learn to ignore, or flag anomalies that nobody can act on. STRATA turns ordinary one-minute trend data into an event log along the control sequence, discovers process models at the device and unit level, and scores each day through eight calibrated, physically interpretable channels against a measured noise floor, so that every alarm names a location, states its evidence and is held to a false-alarm budget. It was built under a falsification protocol: every hypothesis was pre-registered with the observation that would refute it (its falsifier), every result was recomputed independently before it was quoted, and automated tests pin the published numbers and the falsifiers that fired.

On five public LBNL benchmark systems, including the two fan-powered terminal units for which no detection results had been published before, STRATA detects 158 of 175 seeded faults, most on the first day, at 26 false-alarm days in 480 held-out fault-free days (1 to 9.4% per system). Two systems were onboarded blind, by configuration alone. The alarm names the correct terminal unit in 37 of the 50 detections with known ground truth and a wrong one in none, and the faulted subsystem in 83% of all detections. The false-alarm budget holds under injected sensor error, change-of-value logging and schedule shifts (worst 7 of 96 days) and on one measured rooftop unit (3 of 49).

Process mining helps to get detection but is not detecting on its own. The strata and the event abstraction carry the localisation, and invariants that discovery surfaced became several of the rules that detect. Conformance checking of the discovered models, pre-registered as the detector, was refuted: over 199 scenarios on six systems, no fault is caught by the models alone. A principal-component baseline detects three fewer of the 73 scenarios it shares and cannot name the device. A third-party Guideline 36 rule battery, held to the same gate, matches STRATA within one detection on the air handler and detects a third as many faults on the terminal units, so the pre-registered improvement claim over current practice is withdrawn there. The evaluation also produced eight machine-verified errata in the benchmark datasets. Every number regenerates from the public repository.

1 Introduction

Buildings and construction account for 32% of global final energy use and 34% of energy-related carbon dioxide emissions [51]. HVAC accounts for up to half of an individual building’s consumption [8] and for about 30% of all commercial building energy in the United States [18], and 15–30% of that is wasted through faulty operation and poor control [8,27]; correcting the common faults saves 5–30% of whole-building energy [8,43]. The faults persist even where detection is installed. Across

60,000 pieces of equipment in 317 commercial buildings, Crowe and colleagues found 245 faults per building per month and 40% of air handlers in a faulted state on any given day, in buildings that run modern automation systems with alarms [12]. Seven of their ten most common faults are behaviour-based: wrong schedules, unreleased overrides, missed setpoint resets, broken economiser sequences. A threshold alarm or a pre-written rule [44] catches such a fault only if someone wrote that rule in advance [22]. The public benchmarks used in this paper seed component, sensor and control-loop faults and none of that class, so this argument motivates the design and is not tested by it.

Research has gone the other way. Machine learning has been surveyed across 211 HVAC fault-detection studies [8], with accuracies saturated at 97–99% on the standard benchmarks [42, 50] and interpretability added afterwards at the level of input features [11, 14], which tells an operator that a sensor mattered, not what the building should have been doing. Of 65 primary studies audited in 2025, two shared code links, one of them broken, and 72% do not say whether their data is public [36].

What an operator needs is a detector that works from the trend data already recorded, names the unit and subsystem, explains the alarm in the physics of the equipment, is held to a false-alarm rate measured on the building’s own fault-free days, and reaches a new building by configuration rather than new code. Process mining, the one methodology built to model the sequence in which a system moves between states, has not been brought to this problem: to our knowledge no published work applies event-log process mining to building HVAC fault detection. The 211-study survey contains no process-mining entry [8], the reviews of process mining in Industry 4.0 and on sensor data contain no building HVAC application [2, 9], the Lawrence Berkeley National Laboratory taxonomy of fault-detection methods has no such leaf [21], and our own exact-phrase searches of the OpenAlex corpus return none [41].

STRATA applies process mining in stratified form: sensor streams are abstracted into events along ASHRAE Guideline 36 control sequences [3], Petri nets are discovered per device and per unit [52], and the channels that read the log sit beside physics rules and residuals that read the raw signals, each calibrated on its own noise. Our first hypothesis, that the discovered net would itself be the detector, was pre-registered and refuted: a single hand-written rule out-covered the model channel where it was strongest. What settled is the division of labour, and it is a productive one. The strata and the event abstraction give every alarm its location, the frequency and oscillation channels are the only ones that catch the unstable-controller faults, invariants that discovery surfaced became several of the rules that detect, and conformance checking of the discovered nets, the part we expected to detect, contributes no detection of its own. Process mining helps to get detection but is not detecting on its own.

The paper contributes (1) the framework and the detector it yields, evaluated on five public benchmark systems including the two fan-powered terminal units for which no detection results had been published before; (2) what process mining contributes to HVAC fault detection, established under pre-registration: the first such application, whose structure gives the detector its location and several of its rules, and whose conformance channel catches no fault the calibrated channels miss; (3) an evaluation protocol of pre-registration, honoured falsifiers, independent recomputation and full regeneration of every committed result file (artefact); and (4) eight machine-verified errata in the benchmark datasets.

2 Research Aim and Objectives

This research set out to build a fault detector for building HVAC systems that an operator can trust and act on: one that works from the trend data the building already records, explains each alarm in the physics of the equipment, names the unit and the subsystem at fault, is held to a false-alarm rate measured on the building’s own fault-free days, and reaches a new building by configuration rather than new code. Process mining was the means chosen to get there, because it is the one methodology built to model the sequence in which a system moves between states, and no published work had applied it to this problem. The aim therefore carried a second question: what the discovered process models contribute to such a detector, and whether they detect on their own. This work is the first phase of the project proposed under the name ProcessHeal, whose destination is a building that detects, locates, diagnoses and corrects its own faults; STRATA is the detection-and-localisation framework that phase produced, and Section 9 places it on that path. Everything reported here is validated on public benchmark data; deployment on an operating building is the intended next phase, and Section 9 states what it would take.

A building operator today finds a fault by waiting for a complaint or by watching trend graphs for a curve that has drifted from where it usually sits; both find it late and neither names the equipment responsible. STRATA is meant to consume a building’s ordinary one-minute trend data and one configuration file naming its sensors and control rules, learn from a fault-free period what healthy operation looks like, and then, every day, name the days that break a physical relationship the healthy period established: which unit, which relationship, by how much, and how often that channel has cried wolf on healthy days. The output is a work order with a false-alarm budget the operator can hold the system to, and onboarding a new equipment type is a configuration exercise. That intent fixes six objectives, each reported with what was done and where it stands.

1. Abstract events from sensor data along Guideline 36 control sequences, with fault-signature events held out of model discovery. Each day is a case and every event is emitted by a configured rule: three kinds produce the state events model discovery may see, five produce fault-signature events it may not. The same code, with no building-specific additions, ran on seven LBNL datasets; the alphabet wall is enforced by unit tests and the circularity it prevents was measured. Met.
2. Discover models and establish what they contribute. Nets discovered at device, unit and cross-system strata on every system; their contribution measured under pre-registration on 199 scenarios across six systems. Met: the structure, the event log and the invariants they surfaced are what they contribute; the hypothesis that they would also detect by conformance alone is refuted.
3. Detect against a measured noise floor. Eight channels, every threshold from the fault-free year, every false-alarm rate out-of-sample; the budget held under injected field conditions and on one measured building. Met; the statistical repair that made this true cost seven detections, and that is stated.
4. Benchmark against known answers. 158 of 175 seeded scenarios detected at 26 false-alarm days in 480; against an audited PCA baseline, 64 of 73 to its 61; against the open-fdd Guideline 36 battery held to the same gate, within one detection on the air handler and three times the detections on the terminal units. All three parts executed; the improvement claim over current practice is withdrawn on the air handler.

5. Make every claim checkable. More than thirty pre-registered experiments, ten adverse results and four later fired falsifiers published, every headline number regenerated under a regeneration gate, a test suite that fails if a caveat is dropped. Met.
6. Release the work openly. Repository, configurations, artefacts, reproduction path and eight machine-verified dataset errata are public. Met.

What the objectives add up to is a detector that does what the intent asks on the benchmarks it was tested on, a measured and stated answer to the process-mining question, and a protocol that made both checkable. What they do not yet add up to is a field result; Section 9 states what that would take.

3 Related Work

Machine learning for HVAC fault detection. Bi and colleagues survey 211 AI-based HVAC fault-detection studies from 2013 to 2023 [8]. Accuracies on the standard benchmarks have saturated: 98% across thirteen fault types with a random-forest and support-vector pipeline on RP-1312 data [50], 97% on three real buildings with a convolutional-and-LSTM hybrid whose authors name interpretability as the problem they did not solve [42]. The interpretable line pairs a deep model with a post-hoc explanation: Deng and colleagues’ graph network on the LBNL single-duct data reaches 88% at 500 labelled samples with per-sensor importance as the explanation [14]; attention and relevance propagation do the same on chiller data [30, 59]. The explanation is at the level of input features, not of the control sequence the building was supposed to follow. The opportunity is a method whose model is its explanation.

Unsupervised methods and cross-building transfer. Real buildings rarely carry fault labels, so the strongest recent work is self-supervised: a masked-reconstruction Transformer on an unlabelled heat-pump building [1], and El Mokhtari and McArthur’s autoencoders across five fan coil units at Toronto Metropolitan University, the closest precedent for a Canadian university deployment [15]. Each yields a scalar anomaly score. Cross-building transfer in this literature means transferring learned weights by domain adaptation [17, 29]; what STRATA transfers is a configuration file naming sensors and rules, a different operation, and the discovered models themselves did not transfer (Section 7).

Process mining and the empty quadrant. Process mining discovers models from event logs, checks conformance against them and enhances them with further perspectives [52, 53]. Conformance was used directly as a trace-level anomaly detector [7] before neural sequence models overtook it on standard log corpora [38]; our own ablation, which finds no fault detected only by conformance, is consistent with that motivation. In adjacent cyber-physical domains the methodology has succeeded: robotic-arm fault diagnosis at an F1 of 98.9% [57], with alignment features feeding downstream classifiers for gains of 30 to 70 F1 points [58]; industrial IoT sensor validation [24]; industrial control systems [37, 46]; and pasteurisation, where continuous sensor signals are abstracted into activity sequences and checked against a process specification [35]. The common barrier is event abstraction [10, 32]. Building HVAC is an empty quadrant: the 211-study survey has no process-mining entry [8], the 47-study Industry 4.0 review [2] and the 36-study sensor-data review [9] contain no building HVAC application, construction-management process mining treats projects as cases and human tasks as activities [33], LBNL’s fault-detection taxonomy has no such leaf [21], and exact-phrase searches of OpenAlex [41] and the ICPM 2025 programme return none.

Rule-based practice. Fault detection in real buildings is rule-based: commercial platforms such as Automated Logic WebCTRL, installed at the Campus Activity Centre of Thompson Rivers University, ship libraries of over one hundred named faults, and Granderson and colleagues’ survey of fourteen commercial tools confirms the style [21]. LBNL’s automated fault-correction line applies hand-coded correction algorithms through BACnet writeback [31, 43]. Both assume the faults of interest can be enumerated in advance. STRATA is complementary: it learns where the building’s own healthy operation is invariant and holds those invariants at a measured false-alarm budget, and its located, evidenced alarms could in principle feed the correction algorithms. Section 6.2 measures both the strength and the limit of the rule library on the same benchmarks. Hand-designed Petri nets for HVAC [13, 56, 63] are a different use of the formalism: authored by an engineer, with no event log and no discovery, and should not be conflated with the discovered models used here.

4 Method

STRATA (Figure 1; code and artefacts at <https://github.com/iyassh/strata>) consumes one fault-free year of ordinary building-management-system trend data and a single configuration file naming the building’s sensors and control rules. The configuration file is the only site-specific artefact; no code change was needed to onboard any system reported here. The framework is stratified in two senses: events are abstracted from the control sequence at device and unit granularity, so every alarm is born with a location, and the channels that read the resulting log (absence, frequency and the two conformance channels) sit beside physics rules, residuals and an oscillation channel that read the raw signals, each calibrated on its own noise.

Event abstraction and the alphabet wall. Continuous sensor streams become a discrete event log through control-sequence rules grounded in ASHRAE Guideline 36 [3], following Janssen and colleagues’ abstraction methodology [25] and the event-abstraction taxonomy of van Zelst and colleagues [55]. Each day is a case. Every event is emitted by a configured rule of one of eight kinds: occupancy, operating mode and time window produce state events, the only events admitted to model discovery; mismatch, leak, setpoint deviation, paired residual and envelope residual produce fault-signature events, which already encode engineering suspicion and are withheld from discovery. The partition, the alphabet wall, exists because of circularity, the process-mining form of data leakage [61]: a model discovered over an alphabet that names faults will later “detect” them by recognising the project’s own rules. Measured on the single-duct system against the pre-partition prototype, the model channel’s apparent recall rises from 1.8% to 5.2% without the wall, while the deployed detector’s output is unchanged, because every leaked model detection was already a rules detection. The separation is enforced by unit tests: injecting signature events into a log cannot change a day’s fitness.

Discovery at three strata. Petri nets are discovered from the state-event log with the inductive miner [28] in pm4py [6] at three granularities: the device stratum models an actuator’s lifecycle, the unit stratum the air handler’s daily cycle, and a cross-system grammar supports the transfer analysis of Section 7. Whether a net encodes order at all is checked directly by a reversal probe: every trace is reversed and re-scored. The single-duct net’s fitness falls from 0.886 to 0.553 under reversal; the two fan-powered nets do not move (0.953 and 0.902 to 0.903), so their fitness carries no ordering information.

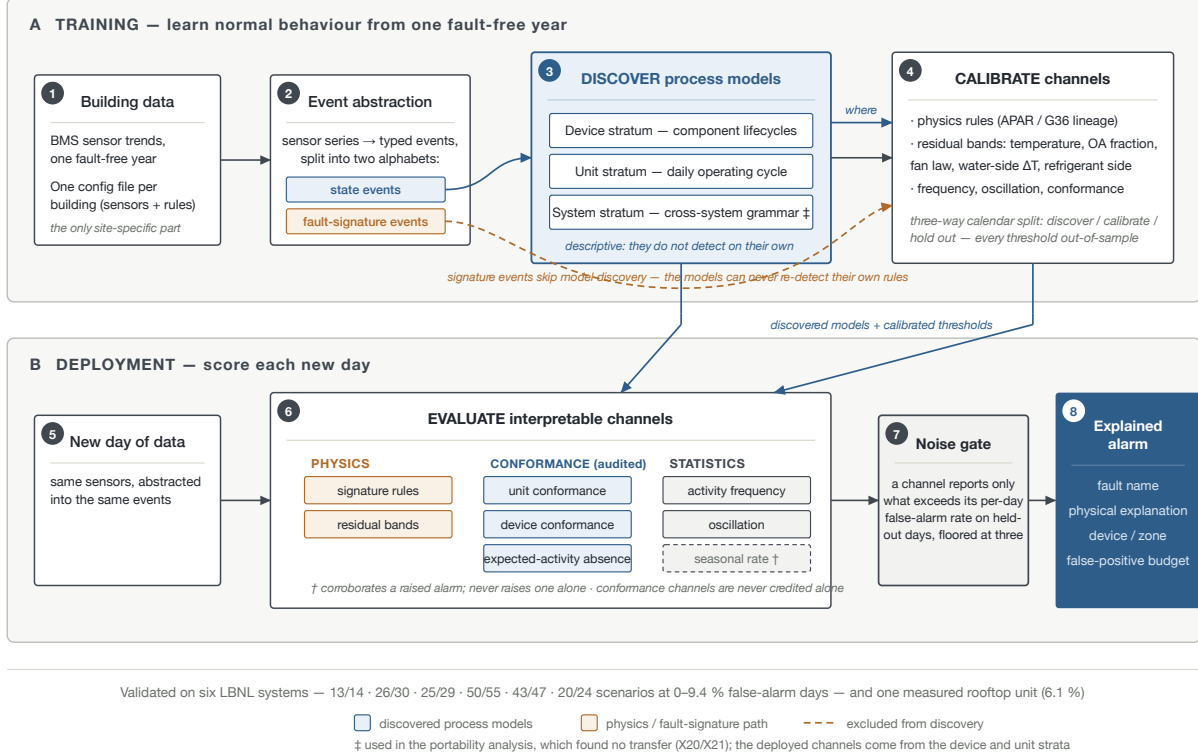


Figure 1: The framework as evaluated. Training (top): one fault-free year and one configuration file; events are split into state and fault-signature alphabets; process models are discovered from state events only and supply the strata and the invariants the channels use, but do not detect on their own; every threshold comes from a three-way calendar split, so every false-alarm rate is out-of-sample. Deployment (bottom): each day is scored by the physics, statistical and conformance channels; a channel reports only what exceeds its own measured noise floor; what survives reaches the operator with a name, a physical explanation, the device and the stated budget.

Calibrated channels and the noise gate. Eight channels score each day independently (Table 1): rules from the APAR and Guideline 36 lineage [44] that name direct contradictions between measurements; residual bands on physical relationships that must hold in a known mode; model and device conformance, the alignment cost [54] against the unit and device nets; absence of a scheduled activity; frequency and oscillation bands on activity counts and direction reversals; and a seasonal rate channel that corroborates but may not alarm alone. Calibration uses the fault-free year only, split three ways by calendar: the last eight days of each month (96 days) are held out and report every false-alarm rate; the eight days before them form a calibration slice for the two conformance thresholds; the band channels learn their extremes on every other day and discovery uses the rest. No threshold sees fault data or the days its false-alarm rate is reported on. A channel then reports only what exceeds its own noise floor, a binomial test at $p < 10^{-3}$ against that channel’s holdout false-alarm rate floored at three days [23]; the rules channel, silent on the whole fault-free year by construction, is held to the same rule of three on the year. Detection thresholds are floored at sensor physics (0.5°F for temperature residuals), which retracted an early result of 21 to 117 detection-days per scenario to at most two. What survives the gate reaches the operator with the implicated device or zone, the evidence, and the channel’s false-alarm budget.

Table 1: The eight channels as deployed. Every threshold is set from the fault-free year alone. The rate channel is advisory: reported, never counted in the detector’s alarm.

Channel	Reads	A day is flagged when	Threshold source
Rules	signature	a control-sequence rule fires during operation	documented bands; rules not silent on the fault-free year are removed at onboarding; gate 3 of 365
Residual	raw signals	a per-mode physical residual leaves its healthy band	[min, max] of training days, widened to a per-rule sensor floor
Model conformance	state	the day’s trace aligns to the unit net below threshold	1 % quantile of fitness on the calibration slice
Device conformance	state	an actuator’s lifecycle trace aligns to its device net below threshold	as above, per device net
Absence	state	a scheduled device produces no lifecycle event on a scheduled day	each device’s healthy silence rate
Frequency	state	a daily activity count leaves its healthy range	[min, max] over training days
Oscillation	raw signals	an actuator’s daily reversal count leaves its healthy range	[min, max] over training days
Rate (advisory)	state	a 30-day active-day rate leaves the range of month-matched healthy windows	training-year windows

5 The Falsification Protocol

An audit of 65 primary machine-learning HVAC fault-detection studies found two that shared code links, one of them broken, and 72% that did not say whether their data was public [36]. A field in that condition needs results that can be checked more than another accuracy figure, so verification was treated as a deliverable. The protocol has three stages and one ledger.

Before a result. Every experiment is pre-registered: the hypothesis and the observation that would refute it are committed to version control before the run [39, 40]. A falsifier that fires is a finding and is published. Ten results went against the project (four refuted hypotheses, one fault-informed rule, three withdrawn claims, one test without attainable power, one transfer test whose pass a discovery-free control reproduced) and four later tests fired their falsifiers; Appendix A lists every experiment with its prediction and outcome. Before any result is computed, a battery of checks runs on the raw data: file hashes, monotonic timestamps, calendar consistency and rotation, Brick coverage [4], and event-log hashes against duplicate scenarios; a healthy-silence gate confirms per configuration that the rules are silent on the fault-free year. Three of the eight dataset errata were found by the gates, the rest by audit, and every one would have silently altered a published result. They are listed in Table 2; the companion paper [47] gives each with its reproducing command.

ID	System	Defect	Consequence
E1	SDAHU	One outdoor-air-bias run shipped four times under different severity labels; the labelled bias never appears in the logged column.	Four labels index one run.
E2	SDAHU	One coil-leakage run shipped four times under different severity labels.	Four labels index one run.
E3	SDAHU	A constant of about 401.86 added to SA_SP in the fault-free file and subtracted from SA_SPSPT in every fault file.	Either column identifies file provenance perfectly; both excluded from all baselines.
E4	SFPU	One file’s calendar is rotated relative to its siblings.	Excluded from family claims.
E5	SDAHU	The fault-free file was simulated on a different configuration branch from every fault file.	One detection was the branch, not the fault: 14 of 14 naive, 13 of 14 adjudicated.
E6	FCU	Two fouling files are byte-identical under two family labels.	Scored once; the fouling family is 11 scenarios.
E7	PFPU, SFPU	Documentation places every fault in the west zone; only the south zone’s columns move.	Zone ground truth established from the data.
E8	DDAHU	Mixing-box statics shift by one tenth of the labelled static-sensor bias.	Severity claims from those labels are off by ten; detections unaffected.

Table 2: Eight machine-verified defects in the LBNL benchmark datasets.

Before a number is quoted. An adversarial audit recomputes the claimed quantity independently from raw data, by an auditor briefed to find it wrong (here, AI coding agents, denied the answer key for the sealed test). It caught a baseline detecting file provenance rather than faults; one single-duct detection that was a dataset artefact, so the score is 14 of 14 naive and 13 of 14 adjudicated; a significance test with an impossible null, retracted; detections below sensor precision; and, twice, a completed comparison against commercial practice, the first execution invalid for three reasons and the repaired one for six adapter defects and one defect of design (Section 6.2). None would have been caught by a test suite, because the code was doing what it was written to do.

After a result. About two hundred tests run on every commit; a regeneration gate re-derives every artefact and diffs it against the stored copy, so a changed value is a defect or a documented decision, never a silent update; every validated dataset is frozen as a fixture. The suite pins published values, fired falsifiers and disclosure obligations: a regeneration that drops a caveat fails the build exactly as a lost feature would. Continuous integration has guarded that a result regenerates [26]; we have not found it reported that it guards that a caveat survives.

What it taught. Sixty-seven numbered lessons carry three patterns: a lesson is not closed until a test enforces it, because documentation decays; every new channel recreated the same missing noise gate until the pipeline enforced it structurally; and simulation flatters, which is why the sensor-precision floor and the noise-floor requirement exist and why the one measured building is reported as a false-alarm rate, not a detection result. The datasets are public by DOI [19, 20], the environment is pinned, seeds are fixed, and a claim-to-artefact table maps every published claim to the artefact that backs it.

6 Results

6.1 Detection and false alarms

STRATA detects 158 of 175 seeded scenarios on the five simulated systems (14 of 14 naive on the single-duct unit, 13 adjudicated, so 157 under the adjudication) at 26 false-alarm days in 480 held-out fault-free days, at a median of one day, and names the implicated terminal unit in 37 of the 50 attributable detections with none wrong (Table 3). Evaluation began with three public LBNL datasets [20]: a single-duct air handler and the parallel and series fan-powered units, the latter two with no previously published detection result; the dual-duct air handler, the fan coil unit and two rooftop units were onboarded afterwards. Scoring follows the day-level, scenario-as-unit conventions of the FDD evaluation literature [16]; every number is scored by the detector’s final calibration, the per-day noise floor of Section 4. Seven datasets went through the pipeline; six are scored for detection (the measured rooftop unit contributes a false-alarm rate only); the five simulated annual systems carry the headline totals; and the three original systems carry the baseline comparisons on their 73 shared scenarios.

System (LBNL)	points	rules (state / sig.)	scored	detected	false alarms	healthy days
SDAHU, single-duct AHU	26	3 / 12	14	14 (13 adj.)	1 / 96	365
PFPU, parallel fan-powered box	109	13 / 45	30	26	8 / 96	365
SFPU, series fan-powered box	109	13 / 49	29	25	9 / 96	365
DDAHU, dual-duct AHU	114	15 / 56	55	50	3 / 96	365
FCU, fan coil unit	29	7 / 19	47	43	5 / 96	365
RTU, simulated rooftop unit	23	3 / 5	24	20	0 / 28	100
RTU, measured rooftop unit (Site 2)	26	2 / 7	1 case	—	3 / 49	182

Table 3: The seven LBNL datasets as scored by the final calibration: points mapped, rules declared, scenarios scored and detected, and the out-of-sample false-alarm count on the held-out fault-free days (the last eight of each month). The measured unit carries one fault case and is used for its false-alarm rate only. With every channel counted, including the advisory rate channel, the single-duct rate would be 15.6%.

Time to detection. The median is one day on all six scored systems; the tails are fouling and charge faults: on the dual-duct unit the moderate and severe waterside cooling fouling at 114 days each, on the fan coil unit the moderate heating waterside fouling at 15 days, on the simulated rooftop unit the 20% refrigerant undercharge at 14 days. Gradually developing faults are detected slowly, the expected cost of calibrating below a noise floor. Pooled recall misleads for a gated detector: on the single-duct coil-bias scenarios the residual is physics-bound only inside a detection window (coil off, zone occupied, sustained 120 minutes) that exists on 137 to 164 days of the year, and within those windows recall is 100% on every scenario; the apparent 39–47% ceiling is window prevalence, not capability.

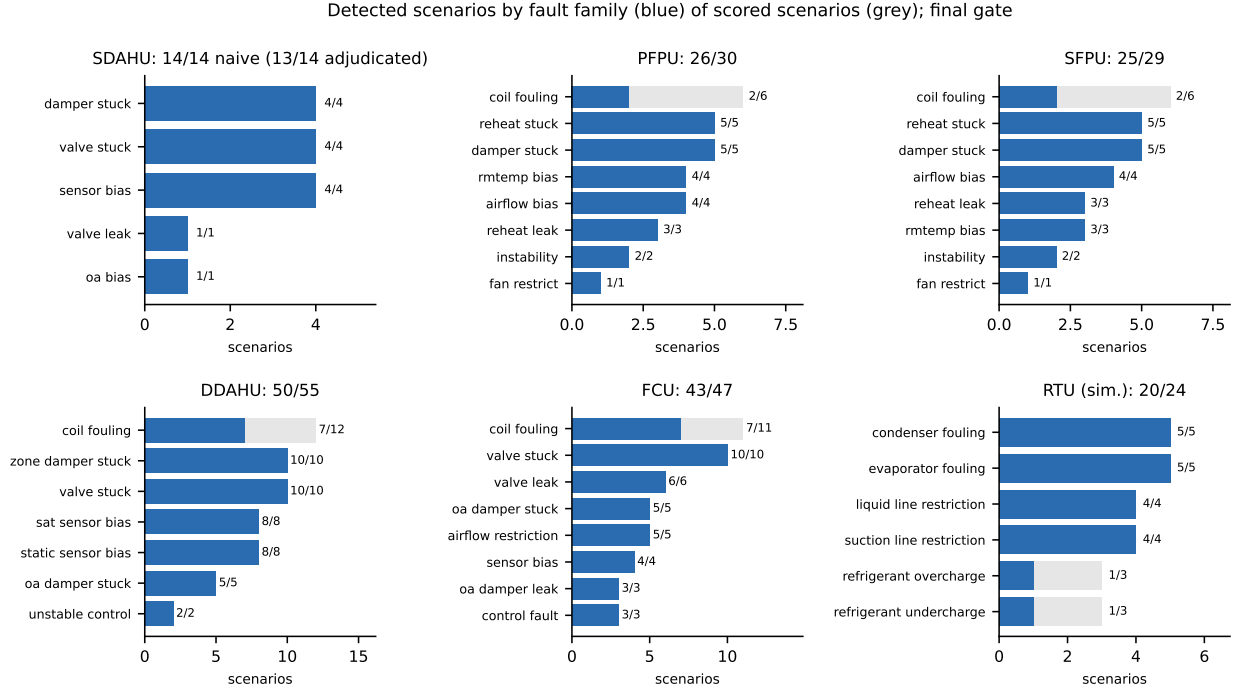


Figure 2: Detected scenarios (blue) of scored scenarios (grey) by fault family on the six scored systems, as scored by the final calibration. Coil fouling is the dominant miss on the four air-side datasets and is caught in full on the simulated rooftop unit, condenser fouling through the refrigerant side and evaporator fouling through the air side (Section 7).

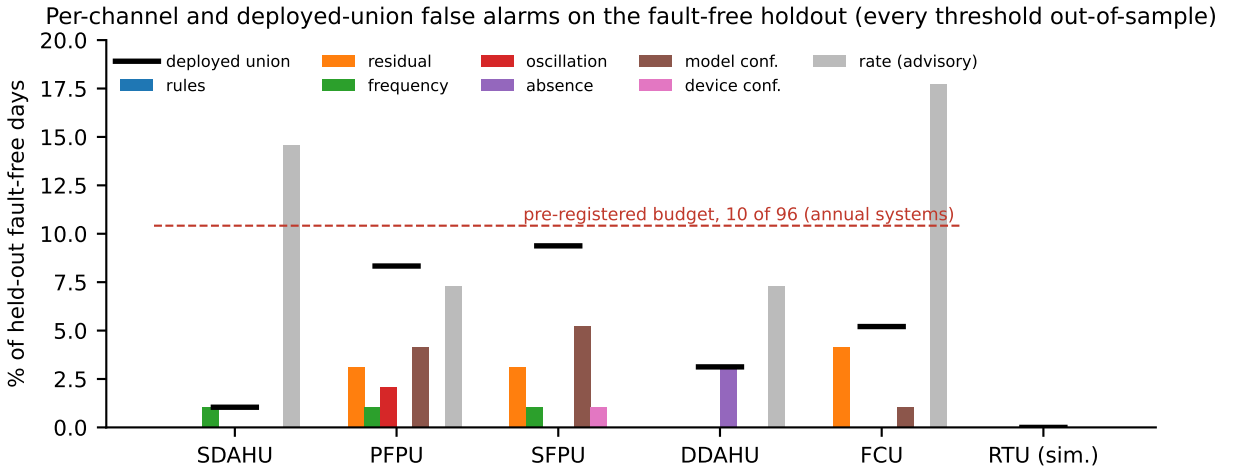


Figure 3: False alarms on the held-out fault-free days, per channel and for the deployed detector as a whole, every threshold out-of-sample. The seasonal rate channel (hatched) is advisory and outside the deployed union; the dashed line is the pre-registered budget of 10 of 96 held-out days for the five annual systems; the simulated rooftop unit's 28-day holdout carries no such budget line.

System	Detected	At 1 day	At 2 days	Remaining TTDs (days)
SDAHU	13 (adjudicated)	9	4	0
PFPU	26	26	0	—
SFPU	25	25	0	—
DDAHU	50	45	0	3, 5, 14, 114, 114
FCU	43	38	1	3, 3, 4, 15
RTU (simulated)	20	19	0	14

Table 4: Time-to-detection distribution behind the one-day median, by system, as scored by the final calibration (from the committed scorecards). The long tails are fouling and charge faults: on the dual-duct unit the moderate and severe waterside cooling fouling (114 days each, carried by the rules channel), on the fan coil unit the moderate heating waterside fouling (15 days, the operating-point indicator of Section 6), and on the simulated rooftop unit the 20% refrigerant undercharge (14 days).

6.2 Two baselines

Against a principal-component baseline. A PCA baseline on squared prediction error [60], fitted to 95% of the fault-free variance with its threshold at the largest training-day score and the provenance-leaking columns excluded, detects 61 of the 73 scenarios it shares with STRATA against STRATA’s 64 (13 / 23 / 25 against 13 / 26 / 25), with fewer false alarms on the two terminal units (1 of 96 each against 8 and 9) and within one day on the air handler. Wilson intervals overlap (64 of 73 is [78.2, 93.4]%, 61 of 73 is [73.4, 90.3]%) and an exact McNemar test on the three discordant scenarios gives $p = 0.25$, an absence of power rather than evidence of superiority [34,62]. The three scenarios only STRATA detects are an unstable VAV damper and the parallel unit’s moderate and severe airside reheat fouling; nine are missed by both, eight of them reheat-coil fouling. What the baseline does not do is name the implicated device or state the evidence behind a flag, which is the difference that matters operationally (Figure 4).

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STRATA alarm -- series fan-powered unit, 2018-01-01 (day 1 of the fault)
what      zone primary-air flow not tracking its setpoint  [channel: rules]
where     terminal unit TU_S, zone S
evidence  measured 715 CFM against a 200 CFM setpoint -- a 515 CFM excess
          sustained for 720 min while occupied; damper position 0.80,
          fault-free same-day value 0.32
budget    rules channel: 0 false-alarm days on the 96-day fault-free holdout
seeded    zone-S damper stuck at 80% -- location correct, symptom correct,
          detected on day 1 of 365
```

Figure 4: A worked alarm, scenario SFPU_VAVDMPRStuck_80%, every value read from the committed artefacts and the raw data. The alarm names the symptom and its location, not the root cause.

Against current practice: the Guideline 36 rule battery. Held to the treatment STRATA gives its own signature rules, a correctly wired third-party Guideline 36 battery (open-fdd 4.4.1 at its installed defaults [3,5]) detects 12 of 14 on the single-duct air handler at one false-alarm day against STRATA’s 13 at one, and 8 of 30 and 9 of 29 on the fan-powered terminal units at no false-alarm day against STRATA’s 26 and 25 at eight and nine. STRATA is within one detection and one

	STRATA	PCA-SPE (strict)
Detected (of 73, adjudicated)	64	61
Per system	13 / 26 / 25	13 / 23 / 25
Holdout false-alarm rate	1.0% [0.2, 5.7] / 8.3% [4.3, 15.6] / 9.4% [5.0, 16.9]	1.3% [0.2, 6.8] / 1.0% [0.2, 5.7] / 1.0% [0.2, 5.7]
Sensor columns mapped (STRATA)	26 of 30 and 109 of 109 (12 and 56 before the coverage audit)	—
Raw daily features consumed (PCA)	—	108 and 424
Implicated device named correctly	37 of 50 (13 unnamed, 0 wrong, 1 indeterminate; the 14 single-duct detections have no device stratum; naive universe)	not applicable
Names the implicated device	yes	no

Table 5: STRATA against the principal-component baseline. Per-system entries are ordered SDAHU / PFPU / SFPU; the counts behind the false-alarm rates are 1/96, 8/96 and 9/96 for STRATA and 1/79, 1/96 and 1/96 for the baseline, with Wilson 95% intervals [62] in brackets. Feature counts are given for the single-duct and the fan-powered systems.

false-alarm day of current practice on the air handler and detects three times as many faults on the terminal units, the difference being per-zone coverage rather than the gate; the pre-registered improvement claim fired its falsifier on every system and is withdrawn. The comparison took three executions: the first was voided for reading water-side temperatures where the standard specifies air-side ones and for grading terminal units with the air-handler rulebook; the second held every prediction and was then voided by two audits that found six wiring defects and a pooled gate that could pass nothing once the battery flagged every healthy day; the third, run under an amendment committed first, is the one quoted. Its caveats: gated per rule, the battery’s fan-powered counts rest on conditions that flag the same days on the fault-free file as on every fault file, so that framing is a property of the held-out floor, not detection; the library’s installed defaults are tighter than the Guideline 36 values its own labels cite; and the terminal rules do not read the building’s own setpoints.

6.3 What process mining contributes

What the discovered models contribute. For each channel a hand-written equivalent was authored and scored. On the unstable-room-loop scenario two such rules reproduce the frequency channel exactly (141 of 141 flagged days on the series unit, 206 of 206 on the parallel unit): the stratified log’s own count bands make explicit rules an expert could have written, had they known where to look. Transplanted without that knowledge the same rules are unusable, firing on 124 fault-free days on the parallel unit and none on the series unit. Some faults need the classical instrument: the oscillation channel alone carries the series unit’s unstable-damper scenario, 349 flagged days against at most two from any rule. The decisive ablation reads the channel recorded as carrying each detection from the committed results files (the scorecards): no scenario on any system is detected only by the two conformance channels. They are credited on none of the single-duct or parallel-unit detections and on 7 of 25 series-unit detections, never alone; removing them leaves every scorecard unchanged and lowers three systems’ false-alarm rates (9 to 4, 8 to 6 and 5 to 4 of

96). The discovered models contribute the strata and the invariants that shaped the rules; they do not contribute detections.

What the event log carries. Day-by-day comparison of the parallel unit’s fault-free year against its scenarios on the state alphabet shows that under waterside reheat-coil fouling the state-event log is identical to healthy to the minute; nine files have that property, among them every fouling miss that remains on the two fan-powered units, and for them no method that reads that log can detect, process mining or otherwise, because at the deployed alphabet the fault lives in heat-transfer magnitudes the abstraction discards. Four pre-registered vocabulary experiments (X12–X15) tested a time-perspective model, a fault-informed coil residual and an enriched alphabet: the enriched frequency channel alone stays in budget and adds one airside-fouling detection, a scenario the PCA baseline also reaches; nothing else new was found, and richer vocabulary bought detections only at false alarms the budget forbids.

What the conformance channel can see. A pre-registered positive control injected order violations into each system’s fault-free holdout traces. The fan coil and dual-duct nets see order (a reversed day is flagged on 55 of 72 and 59 of 83 holdout days, AUC 0.99); the single-duct net separates it on average (AUC 0.91) but no in-budget threshold sees it, because healthy days fit worse than a reversed day; the two fan-powered nets do not move under any perturbation. A pre-registered grid of seven process-mining instruments (inductive miner at two noise settings, token replay, heuristics miner, log skeleton, Declare constraints, temporal profile) detects none of the 33 scenarios the deployed detector then missed, at false-alarm rates of 0 to 9.7%. The conformance null is a property of the state-event log, not of the miner or the measure.

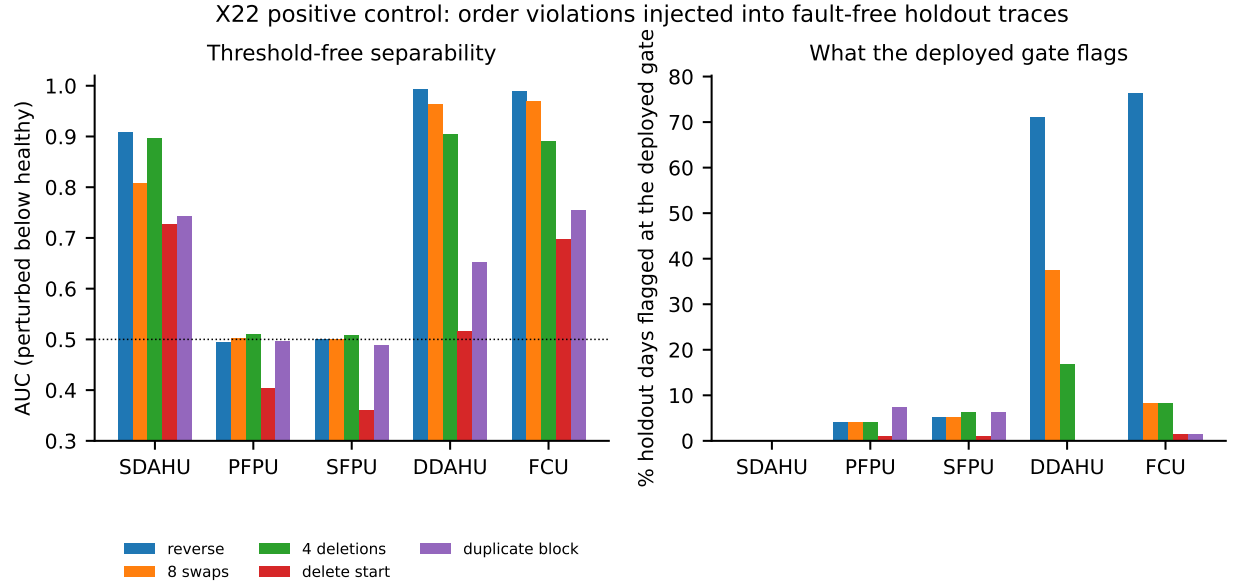


Figure 5: The positive control. Left: threshold-free separability (AUC) of perturbed against unperturbed fault-free holdout days for each injected order violation. Right: the share of those days the deployed gate flags. The fan coil and dual-duct units see order; the single-duct unit separates it on average but not at any in-budget threshold; the fan-powered units’ nets accept any order.

6.4 Localisation, onboarding and coverage

Localisation and portability. On the 51 detections on the two fan-powered systems whose alarms name a device, the alarm names the correct terminal unit in 37, none in 13 and a wrong unit in none, with three wrong units available in every case; because every seeded zone fault is in the same zone, this is reported as specificity, not discrimination. Over all 158 detections the rule carrying the detection names the seeded component exactly in 109 (69%), the right subsystem but a neighbouring component in 22 (83% at subsystem level), the wrong subsystem in 20 (12.7%) and no component in 7; the pre-registered bar for a component-level claim (wrong at most 10%) was missed, so the claim is subsystem-level. Fourteen of the twenty wrong cases are on the single-zone fan coil unit and name the component that responds to the fault rather than the one that caused it. A pre-registered diagnosis layer of signed residual patterns, redundant-copy consensus and a nearest-pattern family vote raised the exact count to 112 (70.9%, 85% at subsystem level) at a 10.1% wrong-subsystem rate and named the family in 85 of 114 resolved cases with 44 unresolved; both its bars were missed and it is reported, not adopted. Portability was measured: the second system type was onboarded by configuration alone in about 1.5 hours by commit timestamps, and no pipeline code changed for any system.

Two systems onboarded blind. The dual-duct air handler (114 points, 55 faults in 15 families, two of them new) was onboarded under a pre-registration written before any data file was opened: 80 mappings, 15 state and 26 signature rules, one healthy-silence iteration, no source change; 45 of 55 detected at 3 false-alarm days in 96, 50 of 55 after the coverage audit. The fan coil unit (29 points, 47 scored faults in 17 families, four of them new) followed under the same form: its first run fired two falsifiers because a scoring rule written for the fan-powered units read this unit's idle setback as a scheduled mode; a two-script repair was pre-registered before it was made, the four earlier scorecards regenerated byte-identically, and the unit scores 41 of 47 at 4 false-alarm days in 96, 43 of 47 after the coverage audit. Nothing on either unit is detected by conformance alone.

The coverage audit. The misses that remained were read as limits of what the datasets record until a read-only scan showed that for six of fourteen the decisive signal was a column no configuration mapped: the configurations had been written from the control sequences, not from an audit of every recorded column. A pre-registered series audited every column of every system, gave each a physically named reader or a written exclusion, and kept every band from the fault-free file with the state alphabet untouched (Table 6). Sixteen scenarios were recovered and none lost for five false-alarm days over 480: airside reheat fouling through the series fan's power per cfm delivered ("same power, less air"), room-temperature biases through the return air falling away from the biased zone, deck static and temperature biases through the unbiased copies the mixing boxes record (282 of 282 days at zero false alarms), and the fan coil's damper leaks through the outdoor-air fraction at a fixed damper position. The seventeen misses that remain are coil fouling whose recorded values sit inside every healthy band on every column. Two disclosures bound the claim: which columns to read was decided after a scan of the fault files, so the readers were fault-informed even though every threshold was blind; and five of the sixteen recoveries sit less than one band width outside the healthy extremes, though all five survive bands ten per cent narrower and four of five survive injected field conditions (reported with the limitations below).

System	columns mapped	detected, before → after	false alarms, before → after	lost
SDAHU	12 → 26 of 30	14 → 14 / 14	1 → 1	0
PFPU	56 → 109 of 109	22 → 26 / 30	7 → 8	0
SFPU	56 → 109 of 109	21 → 25 / 29	6 → 9	0
DDAHU	79 → 114 of 114	45 → 50 / 55	3 → 3	0
FCU	29 → 29 of 29	40 → 43 / 47	4 → 5	0
all	232 → 387 of 391	142 → 158 / 175	21 → 26 of 480	0

Table 6: The coverage audit: sixteen scenarios recovered, none lost, five false-alarm days added over 480 held-out days, the model and device rows identical on every system.

7 What the Evidence Ruled Out, and What It Settled

Thirty pre-registered experiments produced ten results that went against the hypotheses they tested (Table 7). They settle two things the confirmations cannot: what the discovered process models contribute, and how far the method reaches on the data it has. None of them removes a detection from Section 6; each removes a claim about why a detection happens or where the method’s reach ends.

Three of the ten carry the argument. Discovered models do not out-detect hand-written rules: on the series fan-powered unit a stuck zone damper erases the unit’s daily heating rhythm and the discovered net flags the absence, but a one-line rule, occupied workday and heating never active, covers 192 days to the model channel’s 135 at zero false alarms on 365 healthy days. The invariant was surfaced by discovery and the rule written afterwards, which is the division of labour the paper reports. The models’ detections were count deviations in disguise: on alphabets whose events repeat many times a day the nets accept activities in almost any order, and every genuine model-channel detection was extra heating episodes, collapsed fan cycles or a vanished rhythm surfacing through alignment cost, which the frequency channel measures directly and in both directions. And the fouling misses need a different sensor, not one more residual: a pre-registered coil heat-transfer estimate gained nothing, eight of the ten missed fouling scenarios carry no signal in the event log at the deployed alphabet, and on the one dataset that records the refrigerant side the unchanged detector catches every condenser- and evaporator-fouling severity down to 10% at zero false alarms in 28 held-out days, condenser fouling through the condensing temperature’s approach to outdoor air, a point no air handler records.

The transfer test is the protocol’s own lesson. It was pre-registered with its quantity, predictions and falsifiers committed before computation, and it never had power, because the model quantity is constant within a building and three buildings give a minimum attainable p of 0.167; two amendments written under the same discipline did not catch it. The remedy is a pre-registration field naming the unit of replication and the minimum attainable p before any prediction is written. A powered test needs more buildings than three.

8 Limitations

The false-alarm budget held under every stress applied to it: injected sensor error, change-of-value logging and schedule shifts on all five simulated systems cost at most one detection per system and never more than 7 of 96 held-out days, and a calibration ten per cent narrower gained nothing and added thirteen false-alarm days. What remains is this. Every detection result is simulated, and simulation flatters: noise-free years make thresholds cheap and silence easy, the one measured building supports a false-alarm rate (6.1%) and nothing more, and until the deployment phase is

#	What was claimed or hoped	What the test found, and what survived
1	Discovered models out-detect hand-written rules.	A one-line rule covered 192 days against the model channel’s 135 on its best scenario at zero false alarms. Survived: the invariant discovery surfaced became the rule; the models are design-time scaffolding.
2	Structure learned on one building detects faults on another without any configuration for it.	Zero to one detection per building pairing. Survived: portability at configuration level.
3	The conformance channels catch faults no other channel does.	Every genuine model-channel detection was a frequency deviation seen through alignment cost. Survived: the frequency channel, which measures it directly.
4	The single-duct unit scores 14 of 14.	One detection was the dataset’s configuration branch, not the fault. Survived: 13 of 14 adjudicated, quoted with 14 of 14 naive.
5	An early day-level significance test.	Its null was impossible by construction. Survived: a binomial against each channel’s own noise.
6	Calibration tuned at one-minute sampling holds at fifteen minutes, and survives contamination.	28 to 64 spurious healthy days at the coarser rate; one contaminated training day removes two detections. Survived: the healthy-silence gate is re-run at the deployment rate; mild contamination announces itself.
7	Discovery predicts where a rule transfers between buildings.	The unit of analysis was the building, so three buildings gave a minimum p of 0.167 against 0.05: no power, no conclusion. Survived: a pre-registration field naming the unit of replication.
8	A coil-effectiveness residual recovers reheat-coil fouling.	One of twelve fouling scenarios, already detected. Survived: fouling needs a different sensor; the refrigerant-side rooftop unit catches 10 of 10 severities at zero false alarms.
9	The model was starved of vocabulary.	A richer alphabet adds one detection at 7, 18 and 21 false-alarm days in 96. Survived: the enriched frequency channel alone stays in budget, as a candidate, not an adoption.
10	The four-building transfer re-run.	Passes formally ($p = 0.042$), but a discovery-free control passes identically. Survived: no conclusion in either direction; reported as a protocol failure.

Table 7: Ten results that went against the project, each with what survived. Rows 1 to 3, 7 and 8 are the basis of the paper’s statement of what process mining contributes.

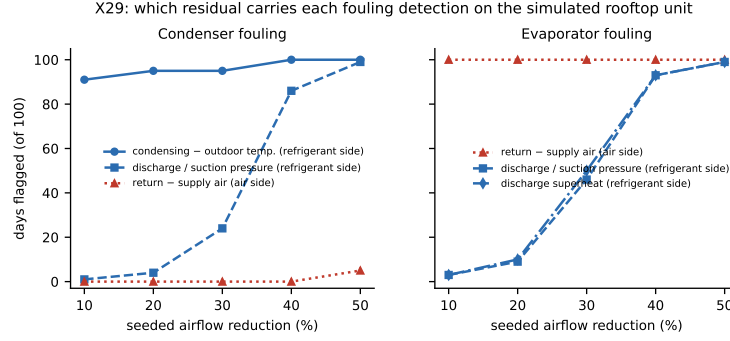


Figure 6: Which residual carries each fouling detection on the simulated rooftop unit, by seeded airflow reduction (days flagged of 100, from the per-residual credit artefact). Condenser fouling is carried by the refrigerant-side condensing-temperature approach at every severity; evaporator fouling is carried by the air-side return-minus-supply residual at every severity, with the refrigerant-side residuals joining from 30 %.

complete the framework is validated in the laboratory, not in the field. Two reporting choices were made after their effect was known and are disclosed: the seasonal rate channel may corroborate but not alarm alone (with every channel counted the single-duct rate would be 15.6%, not 1%), and the conformance thresholds were re-fitted out of sample once the three-way split showed their false-alarm rows had been calibration targets, which moved two budgets up and removed seven residual-only detections. No correction is applied across about 1,600 significance tests, so about 1.6 false cells are expected across the study. The benchmarks seed component, sensor and control-loop faults and no schedule, override or setpoint-reset fault, so the case for sequence-aware detection on that class is motivated, not tested. Gradually developing faults are detected late, the comparison against deployed practice covers three systems and one rule library, and six simulated systems across five equipment classes show configuration-level portability without establishing generality.

9 Conclusion

STRATA works on the benchmarks it was tested on: 158 of 175 seeded faults on five public LBNL systems, most on the first day, inside a stated budget of 26 false-alarm days in 480 held-out fault-free days; the correct terminal unit named in 37 of 50 checkable detections and a wrong one in none, the faulted subsystem in 83% of all detections, and every alarm carrying its evidence. The budget held under injected sensor error, stale logging and schedule shifts on all five systems and on one measured rooftop unit. Against the rule battery commercial building automation systems run today, STRATA is level on the air handler and detects three times as many faults on the terminal units, because it reads the per-zone sensors the rules ignore. Process mining had not previously been applied to this problem; this work made the application and measured what it contributes. Process mining helps to get detection but is not detecting on its own: the strata and the event abstraction give every alarm its location, invariants surfaced by discovery became several of the rules that detect, and conformance checking of the discovered models caught nothing the calibrated channels missed.

To an operator this is a detector whose false-alarm rate is measured on the building’s own healthy days before the first alarm is trusted, whose alarms name the unit, the subsystem and the relationship broken, and whose onboarding is a configuration file rather than a software project:

the second system type took about 1.5 hours by commit timestamps, and the fourth and fifth were onboarded blind with no change to the pipeline code. The proposal that began this work named Thompson Rivers University’s campus energy system, a low-carbon district energy system switched on in September 2026, as the intended testbed [48, 49]; the Campus Activity Centre’s air handler, fan coil units and rooftop units are of the classes tested here and already log the trend data STRATA reads on an Automated Logic system [45]. Deployment there is the next step and is not done. It needs a fault-free period of trend data at the site’s sampling rate, a sensor map written as a configuration with every point mapped or excluded with a reason, and a pass confirming the rules are silent on that period before the detector scores a day; it would deliver a per-day, per-unit alarm stream with a measured false-alarm rate. Scaling follows the same path, which six further datasets have already walked, each at the cost of a configuration, with the silence gate, the regeneration gate and the pre-registration form travelling with it.

The project’s name, ProcessHeal, states the destination: a building that detects, locates, diagnoses and corrects its own faults. Detection, localisation and evidenced alarms exist. A pre-registered diagnosis layer names the fault family from signed residual patterns in three of four resolved cases and resolved the five dual-duct damper cases the carrying rule had misplaced, but missed its bars, and it showed the boundary of passive observation: a stuck actuator and a biased sensor on the same loop leave the same footprint in the recorded points. A self-healing loop needs active tests that perturb a device and watch the response, a witness for cause against response, and an operator study before any automated action [43, 46]. That is a research programme, and the parts that exist are the detector, the localisation, the coverage audit and the protocol under which each further step would be pre-registered. The protocol is the contribution the paper is surest of: pre-registration with binding falsifiers, independent recomputation before any number is quoted, and tests that pin published values, fired refutations and disclosures. Ten results went against the project and are published, eight benchmark defects were documented, and a completed comparison was voided twice before the surviving version was quoted. In a subfield where independent verification is rare [36], a result that has survived a genuine attempt at refutation is worth more than a larger number that has not.

A Every pre-registered experiment

Table 8 lists every pre-registered experiment in the order it was run, with its pre-registered prediction, its outcome and what fired. Each row resolves to a ledger under outputs/ or, where noted in the repository’s research log, to the phase report; amendments are folded into the row they amend.

Table 8: Every pre-registered experiment, its prediction, its outcome and what fired.

ID	Question	Pre-registered prediction	Outcome	Fired / failed
—	Do discovered unit models out-detect a hand-written rule?	The model channel covers more fault days than any matched rule	A one-line matched rule covered 192 days against the model channel’s 135 (178 in the current artefact), at 0 of 365 healthy false alarms	Falsifier fired (matched-rule ablation)
—	Does behavioural structure transfer zero-shot to a new building?	Source-fitted bands detect faults on the target system	0 of 14 single-duct scenarios detected under the confidence-bound gate	Falsifier fired (cold-start detection)

Table 8 (continued)

ID	Question	Pre-registered prediction	Outcome	Fired / failed
—	Does model support predict where a transplanted rule holds? Three amendments.	Spearman $\rho < 0$ with permutation $p < 0.05$ over nine cells	$\rho = -0.69$, $p = 0.200$; the valid null over three buildings has minimum $p = 0.167$	F1 and F2 fired, then both voided; never had power
X5	Are severity ladders monotone in dose?	No ladder is significantly anti-monotone	Series-unit fouling $\rho = +0.87$; no anti-monotone ladder; F-X5.a nearly unfireable at $n = 5$	None
X7	Does healthy silence survive 15-minute resampling?	Healthy signature days and residual coverage are preserved	Healthy years gain 28, 64 and 64 signature days; one residual’s coverage falls 31 to 1	F-X7.a (three systems) and F-X7.b
X8	Does training contamination break extreme-value calibration?	Breakdown at small contamination; mild sources stay visible to rules	One score-carrying contaminated day collapses both negative coil-bias rungs, 164 of 164 to 0	None
X9/ X10	Is the deployed detector robust to sensor jitter and to the holdout split?	Detections change by at most 3; false alarms under 10 %	Counts unchanged at $1 \times$ jitter, within one on the mirrored split; 0 to 4 false alarms of 96	None
X11	Is the single-duct OA-bias detection the fault or the configuration branch?	Five baseline estimators agree; the branch offset explains the flags	Offset $+1.06^\circ\text{F}$; OA-bias drops to 0 flags on 148 window days; 14/14 becomes 13/14	F-X11.d fired on run 1; leg re-specified
X12	Does the model’s time perspective recover missed scenarios?	The duration channel newly detects missed scenarios	11 scenarios significant but 0 newly detected; unique against frequency on 11	None; P3 failed without one
X13	Does a water-side coil-effectiveness residual separate reheat fouling?	At least 6 of the 9 fouling scenarios it counted significant	1 of the 12 fouling files scored significant, already detected elsewhere; nothing newly detected	F-X13.b
X14	Was the discovered model starved of vocabulary?	At least one newly detected scenario inside the false-alarm budget	One newly detected; the single-duct model channel fires on all 14 scenarios instead of none (5 over the three systems before); 7, 18 and 21 false alarms of 96	F-X14.a (budget)
X15	Does the enriched gain survive frequency-only scoring under two splits?	The gain is significant under both splits and inside budget	The same one scenario is significant under both splits; 41 significant against 12 deployed	None
X17	Can a dual-duct air handler be onboarded by configuration alone?	Config-only, gates clean, at least 60 % of scenarios detected	45 of 55 detected (81.8 %) at 3 false alarms of 96; one silence iteration	None; P6 failed (5 fouling, not ≤ 4)
X18	Does the enriched-frequency gain replicate on the dual-duct unit?	A missed fouling scenario becomes significant under both splits	0 newly significant; enriched 22 against deployed 8 on already-detected scenarios	F-X18.b
X19 run 1	Can a fan coil unit be onboarded by configuration alone?	Gates clean, at most 10 false alarms of 96, no conformance-only scenario	All 47 scenarios “detected” at 28 false alarms of 96; 6 conformance-only	F-X19.b and F-X19.d
X19 A1	Does the amended fan coil onboarding hold?	Budget restored and no conformance-only detection	41 of 47 detected at 4 false alarms of 96; 0 conformance-only	None

Table 8 (continued)

ID	Question	Pre-registered prediction	Outcome	Fired / failed
X20	Is model support the log's own heating-day count across four buildings?	The identity holds on every building; $\rho < 0$ at $p < 0.05$	$\rho = -1$, exact one-sided $p = 1/24 = 0.042$; the identity misses by 44 days on the fan coil unit	F-X20.a and F-X20.b
X21	Can a net's reading of a foreign year escape being that log's count?	The bound holds, the tautology persists, the ordering matches the count	The tautology fails on 6 of 12 pairs; the foreign predictor gives $\rho = -0.6$ against the count's -1	F-X21.a; P3 failed without one
X22	Does the conformance channel fire on injected order violations?	Reversal is significant on the single-duct unit	Significant on 2 of 5 systems; the single-duct hit rate is 0 at zero false alarms	None; P1-P4 failed
X24	Does an outdoor-air-fraction residual add detections?	The 20 % damper leak is detected, budget held, nothing lost	Fan coil 41 to 42 of 47 at 4 false alarms of 96; no system loses a detection	None
X25 steps 1-2	Are the conformance thresholds calibrated out of sample?	Rows out of sample, budget ≤ 10 of 96, at most 3 verdict flips	All held at 0 verdict flips; the in-sample rows had understated false alarms by 3 and 4 days	None
X25 step 3	Does a $\max(\text{fp}, 3)/n$ floor on the pooled denominator keep verdicts?	Counts within 2 and at most 3 verdict flips	Held at 2 verdict flips; one dual-duct scenario gained, one series-unit scenario lost	None
X25 step 4	Does a per-day residual null keep the detections?	Each system's count stays within 2	The dual-duct unit loses 3 detections (the series unit 2, the parallel and fan coil units 1 and 2) and 8 verdicts flip across the five systems; budgets unchanged (step-level artefact, before the fan-law channel; the final regeneration, x25_statistical_repair.json, nets seven lost and one gained on the dual-duct unit, the count Section 8 quotes, and fires F-X25.c on the series unit)	F-X25.c
X26 (+A1)	Does the false-alarm budget hold on a real building?	At most 3 silence iterations, false alarms under 10 %, the fault case flagged	6.1 % false-alarm days with every channel in budget; the 40 % undercharge case is not flagged	None; both case predictions failed
X27	Does a fan-law virtual static recover static-pressure biases?	At least 7 of 8 static-bias scenarios detected	6 of 8 detected; budget held and nothing lost	F-X27.b
X28	Does a water-side coil UA channel see the simulated fouling?	At least 3 dual-duct and 2 fan coil fouling scenarios gained	0 gained on either system; run 1 also broke the fan coil budget	F-X28.b; run 1 also F-X28.a
X29	Is fouling visible when the refrigerant side is recorded?	At least 3 of 5 condenser and 3 of 5 evaporator severities detected, monotone	All 10 fouling severities detected and monotone at 0 false alarms of 28	None
X30	Does any other process-mining instrument see the deployed misses?	No cell detects a miss within 3 false alarms	33 cells against 33 missed scenarios, 0 detections; worst cell 7 of 72 (9.7 %)	F-X30.b (two cells not evaluated)

Table 8 (continued)

ID	Question	Pre-registered prediction	Outcome	Fired / failed
X31 (+A1)	Does the detector survive field noise, change-of-value logging and schedule shifts?	Budget held everywhere; the schedule arm within ± 1 detection	Budgets held (run before the coverage audit); the dual-duct schedule arm reaches 48 against the clean arm's 45	None; P3 and the A1 prediction failed
X32	Does a healthy time-of-day envelope per signal, with departures as events, carry detection?	Envelope counts inside the budget; the envelope does not see fouling; order among deviation events adds nothing beyond their counts	Withdrawn before any computation: a read-only design scan showed a third to a half of signals leaving the envelope on healthy holdout days (9–19 % of healthy days flagged); no artefact	None; withdrawn unrun, replaced by X33
X33a-e	Does mapping every recorded column recover the misses? (one system at a time)	No loss; budget $\leq$ clean + 3; named recoveries per system	142 $\rightarrow$ 158 of 175, none lost, 21 $\rightarrow$ 26 false-alarm days of 480, model and device rows identical; PFPU and DDAHU fouling recovered against prediction	None; three predictions failed upward, one (PFPU waterside) downward
X34	Does the carrying rule name the seeded component?	Exact ≥ 60 %; exact or subsystem ≥ 85 %; wrong subsystem ≤ 10 %	Exact 109 of 158 (69 %), subsystem 22, wrong 20 (12.7 %), unnamed 7	F-X34.a; the component-level claim withdrawn
X35	Do the field conditions hold on the post-coverage configurations of all five systems?	Budget $\leq$ clean + 3; noise and COV within -4 ; at least two edge recoveries lost under noise	Budget worst 7 of 96; at most one detection lost per system and condition; one edge recovery lost; clean arm equals the scorecard	None; P3 failed
X37 (3 runs)	Do signed patterns and redundant-pair consensus diagnose family and component?	Family ≥ 80 % with ≤ 15 % unresolved; wrong subsystem < 10 %	Family 74.6 % with 27.8 % unresolved; wrong subsystem 10.1 %; the five dual-duct damper cases resolved	F-X37.a, F-X37.b
X36	Do percentile bands keep the edge recoveries and stay in budget?	False alarms rise ≤ 3 ; the five edge recoveries kept; ≥ 2 new detections	Nothing gained or lost; all five kept; false alarms +4 on both fan-powered units	F-X36.a
X38 (+A1)	Does the repaired Guideline 36 battery match STRATA under the same gate?	STRATA at least as many at lower false alarms (three framings after A1)	Silent-rule battery 12 of 14 at 1 vs 13 at 1; 8 of 30 and 9 of 29 at 0 vs 26 and 25 at 8 and 9	F-X38.a under G2, G3 and S

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